

ClaimLens — Dataset Details

Dataset Reference & Preprocessing Specification

1. Dataset Overview

- **Dataset Name:** Free Synthetic Insurance Claims 100M
- **Primary Hugging Face Source:** [ziadatalabs/FreeInsuranceClaims100M](https://huggingface.co/datasets/ziadatalabs/FreeInsuranceClaims100M)
- **Alternate Source:** Kaggle Insurance Claims Fraud Data
- **Format:** Apache Parquet (100 row groups, ~2.35 GB compressed) & sample CSV
- **Total Records:** 100,000,000 synthetic property & casualty insurance records
- **Coverage Lines:** Auto, Home, Renters, Property, and Business Insurance

2. Schema & 13 Core Fields Specification

Field Name	Data Type	Description	Role in Solution
claim_id	String	Unique claim identifier (e.g., CLM-0000000001)	Document ID in Vector DB
policy_id	String	Unique policy contract identifier	Policyholder linking
claim_type	String	Coverage line (Auto, Home, Renters, Business)	Metadata filter & ML feature
state	String	2-letter US State postal code	Geographic risk & filtering
policyholder_tenure_years	Float	Policyholder tenure duration with carrier	Fraud risk & triage scoring
previous_claims_count	Integer	Number of previous claims filed	Loss frequency feature
incident_date	Date	Date when the loss/damage occurred	Timeline validation
claim_filed_date	Date	Date claim was officially submitted	Filing latency indicator
claim_amount	Float	Total claimed loss amount in USD	Regression target & exposure
deductible	Float	Applicable policy deductible in USD	Net payout calculation
claim_status	String	Open, Approved, Closed, Denied, Under Invest.	Historical outcome precedent
days_to_resolution	Float	Turnaround days from filing to closing	Regression target (duration)
is_fraud_flagged_ground_truth	Boolean	Ground truth indicator for suspected fraud	Classification target (Fraud)

3. Data Preprocessing & Sampling Pipeline

- **Parquet Chunk Streaming:** PyArrow / Fsspec streams row group 0 directly from Hugging Face, extracting a 10,000-record sample for model training, with 1,500 candidates indexed into the active in-memory vector store for sub-2ms dot-product calculations.
- **Handling Missing Values:** Open/pending claims have null days_to_resolution; these are preserved for open status queries and imputed during ML training on closed claims.
- **Text Narrative Synthesis:** Each row is structured into a natural language description chunk capturing all 13 attributes for dense embedding indexing.
- **Feature Engineering:** Derived filing_delay_days (difference between filing date and incident date) and claim_to_deductible_ratio for enhanced fraud risk prediction.